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# 2026 ECMWF-ESA workshop report: current status, progress and opportunities in machine learning for Earth system observation and prediction

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The 5th ECMWF-ESA Machine Learning Workshop (Bologna, Italy; 13–17 April 2026) convened the Earth System Observation and Prediction (ESOP) community to assess progress and new directions in the general field of the application of Machine Learning to ESOP and the specific thematic areas of Machine Learning (ML) for Digital Twins, hybrid ML-physics systems, ML for Earth observation (EO), end-to-end Data Assimilation and Prediction and high-performance computing. The 5-day workshop was held in a hybrid format at the *Tecnopolo Data Manifattura* (DAMA) in Bologna (IT) with an interactive online component, featuring 57 expert talks and 88 poster presentations. The workshop emphasised both the increasing scale and sophistication of ML for ESOP—including potential end-to-end workflows for future operations—and the parallel shift from CPU-centric to GPU/TPU-centric computing, with additional paradigms (quantum, edge, neuromorphic) emerging. Building on the reporting structure of the previous workshop report (ref. 1), this document summarises outcomes across the five thematic areas covering current topics and emerging trends. All the workshop presentations and recordings are available online at <https://ecmwfevents.com/i/5th-ecmwf-esa-machine-learning-workshop/public/agenda>.

The fifth edition of the ECMWF-ESA Workshop on Machine Learning for Earth Observation and Prediction (ML4ESOP) took place from 13 to 17

April 2026 at the *Tecnopolo Data Manifattura* (DAMA) in Bologna, which hosts the ECMWF Data Centre and other major High-Performance Computing (HPC) systems. As in previous editions (refs. 1–4), attendance was high (215 in person and 258 online registered participants) with 57 talks and 88 poster presentations. These numbers confirm the broad interest in ML applications for Earth system science and the central role that the ECMWF-ESA workshop series has acquired as a reference meeting and discussion venue for the ESOP community.

ML is increasingly embedded across ESOP workflows—from EO retrievals and feature detection, through surrogate modelling and down-scaling, to hybrid augmentation of physics-based Numerical Weather Prediction (NWP) and climate systems and data assimilation (DA), and emerging end-to-end approaches spanning observations to forecasts. To help make sense of this broad and rapid evolution of the field, two leading experts were invited to provide overviews of the state-of-the-art in their domain of expertise, as well as discuss current developments and challenges.

Prof. Rossella Arcucci (Imperial College London—United Kingdom [UK]) in her talk “AI for Humanity” highlighted the capabilities of AI-enhanced data assimilation and prediction systems to mitigate the impact of natural hazards such as floods and wildfires as well as broader environmental challenges including greenhouse gases emissions. Her vision emphasised the synergy between DA and ML to extract the maximum amount of actionable information from the rapidly growing volume of observational data, including conventional observations, Earth observation satellite measurements and emerging social data streams.

Prof. Marc Bocquet (ENPC—France) also framed ML progress through the DA lenses. A growing literature now exists on how to incorporate ML techniques in the standard DA workflow to improve both efficiency and performance. At the same time, the new area of Data Assimilation Networks (DAN) and related techniques are receiving increased attention as a potential way to replace the computationally heavy analysis step with a ML system. In both areas theoretical understanding appears mature enough to go beyond proof of concept work and allow testing in more realistic environments.

This edition of the workshop was structured around five thematic areas: (TA1) ML for Digital Twins of the Earth system; (TA2) hybrid ML-physics based systems for DA and Weather and Climate prediction; (TA3) ML applications for Earth system observations; (TA4) end-to-end ML systems for DA and Weather and Climate prediction; (TA5) high performance and new computing technologies for ML applications in ESOP. The following TA sections summarise the workshop’s key messages on the

current status, progress and opportunities obtained from numerous presentations and group discussions.

#### TA1: ML for digital twin of the Earth system

**Current ML applications.** The working group for TA1 was chaired by Rochelle Schneider (ESA—UK) and Mounia El Baz (ESA—Italy). This session examined the rapid expansion of digital-twin-aligned ML across technologies and domains, spanning ML-based modelling approaches, cross-domain transfer, robustness evaluation, and operational deployment.

Foundation-model approaches and generative AI-based emulators have emerged as prevailing methodological frameworks, as exemplified by the *WeatherGenerator* Project (Ilaria Luise, ECMWF—UK). This trajectory was already evident at the previous edition of this workshop, when these models were in their early stages of development. Since then, they have not only transformed the landscape of climate and weather prediction but have also demonstrated clear potential for adaptation to other domains. Instances were presented of these models being transferred from weather prediction to ocean forecasting (Stefano Campanella, National Institute of Oceanography and Applied Geophysics—Italy), or evaluated for regional performance (e.g. Balu Shaharica, Indian Institute of Technology (IIT) Roorkee—India). Complementing these modelling efforts, grounding simulations in satellite observations emerges as a natural and necessary step, and self-supervised ML can offer a promising avenue for bridging km-scale atmospheric models with observational data (Dwaipayan Chatterjee, Institute for Meteorology and Climate Research, Karlsruhe Institute for Technology—India).

As AI/ML models become increasingly central to digital twin systems, questions of evaluation and trustworthiness have grown in importance. These were critically evaluated by comparing them against physics-based models and assessing their ability to generalise to out-of-distribution extreme events. While the limitations of AI models in this area have been extensively documented and are well known, they were shown to produce events with return periods exceeding their training duration, suggesting a capacity for extrapolation into the tail, although physics-based models can still sample further into the tail, producing even rarer events (Amaury Lancelin, Réseau de Transport d'Électricité (RTE)—France). Additionally, through the Systematic Handling of Real-world Uncertainty for Geospatial Foundation Models framework (SHRUG-FM), presented by Joppe Massant (Ghent University—Belgium), uncertainty quantification was also explored as a means to guide end users by providing insight into the sources and extent of model uncertainty.

These ML advances for digital twin systems have translated into a diverse range of applications, spanning both established and emerging fields. Examples include climate-related health risk assessment through urban heat islands and air pollution monitoring (Ana Oliveira, CoLAB + Atlantic—Portugal), sea ice modelling via generative flow models (Tobias Finn, ENPC—France) or using emulators (Flavia Porro, University of Bologna—Italy), coastal ecosystem monitoring (Oscar Leung, The University of Hong Kong (HKU); Ye Ha Kim, University College London—UK), and thermal mapping for urban land cover planning (Hugo Poupard, Murmuration SAS—France). The session also highlighted under-explored applications in weather prediction, including turbulent atmospheric variables which are critical to aviation, laser satellite communication, and optical astronomy, and demonstrated that their vertical profiles can be effectively downscaled using deep learning (Maximilian Pierzyna, Delft University of Technology—The Netherlands). Many ML applications are already moving towards operational deployment, which has emerged as a central theme in the session presentations and in the group discussion, as illustrated by initiatives such as: (i) *Bris* (Norwegian word for “light wind”), a weather

forecasting model based on graph neural networks (GNNs) with an innovative stretched-grid architecture and (ii) *Forecast-in-a-Box*, data-driven forecast models and the entire prediction end-to-end workflow into a portable, self-contained unit (both presented by Lene Østvand, Norwegian Meteorological Institute (MET—Norway)).

**Limitations, opportunities, and challenges.** Along with the current rapid progress in embedding ML applications within digital twin systems, some challenges have emerged. *Computing resources* are a known topic that still faces constraints. While prototypes start small, scaling to larger, more competitive models often requires substantial compute and human resources, enabling well-resourced organisations to iterate faster compared to smaller research groups. *Operationalisation* also faces as many engineering challenges as modelling ones: increasing architectural complexity may improve benchmarks but can reduce trustworthiness and hinder deployment, particularly for research groups that often lack the engineering capacity to deploy operational systems. During the group discussion, the workshop panellists also emphasised that *user adoption* presents a significant challenge, as many stakeholders prefer simplified products over complex, high-dimensional fields. This highlights a trade-off between producing user-friendly outputs and maintaining the level of model complexity required to adequately capture rare, high-impact events. A further challenge concerned the coupling of different digital twin components (land, ocean, atmosphere...), which in ML systems are often treated as separate entities. While some form of coupling is frequently present in numerical models, ML systems do not often mirror this, raising questions about where coupling interfaces can be defined and how to prevent divergence between interlinked components.

**Future directions.** The challenges discussed above also bring opportunities for the field. Model development with greater emphasis on uncertainty quantification, explainability, interpretability, and trustworthiness emerges as a natural next step from current efforts. Interoperability between digital twin components calls for the development of coupling strategies and shared standards for data harmonisation and interface definitions, enabling component reuse across projects. On the user side, investment in human-centred delivery mechanisms, including natural-language interfaces, lightweight agents, and task-specific tools, could help bridge the gap between complex back-end models and stakeholder needs. Finally, the growing availability of kilometre-scale simulations and digital twin outputs opens new avenues for training next-generation ML emulators, including higher resolution vertical profiles and the niche variables that current datasets underrepresent. In parallel, the resource demands of increasingly large models are likely to drive interest in efficient alternatives like smaller, optimised architectures that maintain competitive performance while remaining accessible to a broader user and research community. More broadly, the adoption of AI emulators and geospatial foundation models shows no signs of slowing in the weather and climate communities, and their continued development and diversification across domains will likely remain a defining trend in digital twins.

**TA2: hybrid ML-physics based systems for data assimilation and weather and climate prediction.** The working group was chaired by Rossella Arcucci (Imperial College London—UK), Marc Bocquet (ENPC—France), and Marcin Chrust (ECMWF—UK). The focus of the session was on using hybrid ML-physics approaches for DA and weather and climate prediction, including methods that combine physical models with learnable statistical or machine learning components to improve

forecasting systems, model representation, uncertainty quantification and the assimilation of observations.

**Current ML applications.** Recent work demonstrates growing maturity of hybrid ML-physics approaches embedded within the DA cycle, reflecting a consensus that DA remains essential for providing physically consistent initial conditions and uncertainty characterisation. Keynote perspectives (Steve Penny, Sofar Ocean—USA) emphasised that many ML Weather Prediction (MLWP) systems still depend on NWP inputs, and that end-to-end approaches currently expose important limitations, motivating tighter integration of physics, DA and ML rather than replacement.

A prominent advance is leveraging the DA cycle of NWP to learn model error. Contributions on the ECMWF Integrated Forecasting System (IFS) described neural model-error corrections trained offline from operational increments and refined through online learning formulated as weak-constraint 4D-Var, with neural parameters of the IFS correction included in the control vector (Alban Farchi, ECMWF—UK). Extensions relying on the IFS ensemble DA architecture (“batched” learning) aim to improve robustness by exposing the model to longer and diverse DA windows.

A second line of progress concerns scale-selective coupling of MLWP and NWP via spectral nudging, exploiting ML skill (possibly learned using *adjusted* mean square error metrics) at large scales while retaining the physical model for finer scales and complete variable sets. Results from Leo Separovic (Environment and Climate Change Canada) and Vincent Chabot (Météo-France) reported positive impacts in short-range case studies and surface-focused metrics, while highlighting sensitivity to truncation choices and fine-tuning strategy.

A third area targets flow-dependent uncertainty. Approaches were presented to emulate computationally expensive ensemble-derived DA statistics using ML, with the goal of integrating ensemble DA (EDA) emulators operationally (Wei Pan, ECMWF—UK). In parallel, ML-driven background-error covariance modelling was proposed to support emerging sub-kilometre DA regimes where classical assumptions break and full ensemble approaches become prohibitively expensive (Ravi Shankar Nemani, University of Reading—UK).

Finally, several contributions explored broader aspects of hybrid DA–ML integration, including development of hybrid physics-data-driven models for sea ice prediction (Giovanni De Cillis, Univ. of Bologna—Italy); rapid update, probabilistic hybrid nowcasting systems for precipitation (Simon De Kock, Vrije Universiteit Brussel—Belgium) and multi-hazard forecasting (Claudiu Adam, National Meteorological Administration of Romania—Romania), model error attribution (Yuiko Ichikawa, University of Cambridge—UK), reinforcing the emerging view of hybrid ML–physics approaches as a unifying framework for next-generation DA and prediction systems.

**Limitations, opportunities, and challenges.** Across methods, a recurring challenge is the generalisation of classical DA methods, the adaptation to non-stationary weather regimes, and the separation of physical scales within these schemes. The first two challenges were prominent for online model-error learning in sequential DA cycles. For nudging-based hybrids, deterministic ML training losses can induce excessive smoothing at less predictable scales, limiting benefits at short lead times. Error-statistics emulation, such as ensemble member ML-based generation, faces intrinsic limits from information loss when approximating high-dimensional DA with reduced ensembles. Surface and coupled-DA applications highlight continuing challenges from land-surface

complexity and radiative transfer uncertainty (Patricia de Rosnay, ECMWF—UK).

**Future directions.** The group discussion points to a future where hybrid ML-physics becomes core DA infrastructure and defines it as the logical evolution of the current infrastructures. The emerging vision is a synergistic DA–ML framework, where DA ensures physical consistency and uncertainty quantification, while ML enables faster, more adaptive use of complex and heterogeneous observations. In line with the active research examples previously given, priority development directions include: (i) Operationalisation of DA-native learning of model error and parameters; (ii) refining scale-selective ML–NWP coupling with loss functions and temporal resolutions aligned to DA needs; (iii) advancing multimodal data assimilation by using ML to integrate additional and heterogeneous observation sources while preserving physical consistency; and (iv) operational deployment of ML-augmented DA components, covariances, ensembles and observation exploitation, to extend capability and NWP skills at manageable computational cost.

**TA3: ML applications for Earth system observations.** This session was chaired by Maryam Pourshamsi (ESA—The Netherlands) and Chiara Maria Cocchiara (ESA—Italy). The session addressed current developments in end-to-end ML approaches across Earth observation domains, with emphasis on multi-sensor integration, data-driven inference, and emerging links between observation and modelling systems.

**Current ML applications.** The presentations made during this session can be grouped into three main themes: (i) hazard monitoring and environmental intelligence, (ii) representation learning and data fusion for geophysical retrievals, and (iii) physics-informed and hybrid modelling approaches. For example, related to the use of ML for detecting, segmenting, and monitoring environmental hazards and high-impact phenomena, Dohung Kim (UNICEF) illustrated how satellite data combined with AI can support climate hazard monitoring for children and vulnerable populations. Ana Maria Pelin (TU Delft, the Netherlands) presented methods for detecting and segmenting atmospheric features using geostationary satellite data. Matteo Broccoli (Centro Euro-Mediterraneo sui Cambiamenti Climatici, CMCC—Italy) showed how learning-based approaches can extract mesoscale ocean structures from satellite altimetry, while Ben Pickering (Wx Labs Ltd and NCAS—UK) demonstrated ground-based ML-enabled sensing for high-frequency atmospheric observations. Results from these contributions collectively show that ML-based feature detection and sensing approaches are becoming robust across multiple domains of the Earth system, enabling more timely identification of atmospheric and oceanic phenomena and supporting both operational monitoring and impact-oriented applications.

The second group of contributions focused on learning scalable and transferable representations from multi-sensor Earth observation data. Yufeng Ho (OpenGeoHub, the Netherlands) introduced representation learning approaches for retrieving geophysical variables from heterogeneous EO datasets for biomass estimation. Khunsa Fatima (University of Leicester, UK) presented multimodal classification frameworks for land surface and ecosystem monitoring. Peter Weston (ECMWF—UK) discussed ML-based observation operators for land-surface data assimilation. Results from these studies indicate that representation learning and multimodal fusion methods are increasingly enabling consistent retrieval and classification of geophysical variables across diverse sensors, improving scalability and strengthening links between observations and data assimilation systems.

The final theme captured advances in hybrid and physically informed machine learning. Enno Tiemann (OHB Digital Connect and TUM—Germany) explored domain adaptation techniques for transferring learning from synthetic to real observations. Simona Cariello (INGV-OE—Italy) presented generative modelling approaches for reconstructing geophysical fields under sparse observational coverage. Simone Raciti (University of Bologna—Italy) discussed neural emulation of radiative transfer processes, and Ruben Cartuyvels (ESA—Italy) presented a data-driven reanalysis of Amazon water levels using deep learning. Results from these contributions demonstrate that physics-informed, generative, and hybrid modelling approaches are increasingly capable of addressing data sparsity, improving physical consistency, and accelerating computationally expensive components of Earth system modelling.

Overall, the session demonstrated the breadth of ML applications across Earth system science, ranging from hazard monitoring and environmental intelligence, to physically informed modelling and data assimilation, highlighting a clear trend towards integrated, multi-sensor, and hybrid learning frameworks.

**Limitations, opportunities, and challenges.** The discussion highlighted several key limitations that currently constrain the scalability and operational uptake of ML methods in Earth system applications. A central challenge is the scarcity and heterogeneity of high-quality labelled datasets, particularly for globally consistent training and validation across diverse geophysical regimes and rare/extreme events. This reinforces the need for coordinated community efforts in dataset curation and benchmarking.

A further challenge lies in the so-called “data plumbing” gap. Considerable effort is still required to transform heterogeneous EO data into ML-ready formats and to integrate ML components into existing numerical weather prediction and data assimilation systems, which are often based on legacy infrastructures and programming environments. This creates friction between modern ML ecosystems and operational Earth system modelling frameworks.

The discussion also identified methodological challenges in differentiable modelling and learned observation operators, where issues such as unstable or poorly constrained gradients can limit robustness. Similarly, emulation of high-dimensional forward operators remains difficult due to the complexity and variability of atmospheric states and surface conditions.

From a broader perspective, panellists noted the need to balance openness and accessibility of data with governance requirements, particularly for sensitive datasets. At the same time, the importance of making EO data more “ML-ready” through improved standardisation, metadata consistency, and harmonised data structures was strongly emphasised.

**Future directions.** The group discussion pointed towards several directions for future development. A key priority is the establishment of large-scale, trusted benchmark datasets that support supervised and self-supervised learning across a wide range of Earth system processes, including rare and extreme events. A second major direction is the systematic design of ML-ready Earth observation data products. This includes improved data formatting, temporal and spatial harmonisation, and tools that facilitate multi-sensor integration and large-scale training workflows. A third priority is the closer integration of ML with data assimilation and numerical modelling systems. In particular, learned observation operators and differentiable forward models were identified as promising pathways, provided that issues related to gradient stability, physical consistency, and uncertainty quantification are addressed.

Finally, the discussion highlighted the importance of open access to data and ML tools, including dataset versioning for training and evaluation workflows, consistent metadata and documentation for multi-sensor products, and maintained code repositories for reproducible ML pipelines. This was particularly relevant in the context of collaborative labelling efforts, cross-dataset representation learning, and the increasing use of ML models in operational Earth observation workflows, where reproducibility and traceability are critical for long-term usability.

**TA4: end-to-end ML systems for data assimilation and weather and climate prediction.** This session was chaired by Massimo Bonavita (ECMWF—UK) and Alberto Carrassi (University of Bologna—Italy). The session addressed current developments in End-to-end ML systems for DA, Weather and Climate Prediction and multidomain MLESOP.

**Current ML application.** This session covered observation-centric prediction, DA-aware end-to-end frameworks, expansion from atmospheric forecasting to wider Earth-system applications, interpretability, and probabilistic regional and extreme-event forecasting. Eulalie Boucher (ECMWF—UK) framed direct observation-to-forecast learning through GraphDOP/AI-DOP as a possible pure-ML counterpart to analysis-based workflows, aiming to learn directly from satellite and in-situ observations rather than from reanalysis initial states; Azadeh Gholoubi (NOAA—USA) presented OCELOT, an observation-centric system that forecasts directly in observation space and uses autograd-based forecast sensitivity to observations to diagnose beneficial and detrimental channels. Judongyang Zhou (Imperial College London, UK) explicitly linked end-to-end ML with variational DA for atmospheric composition, combining a data-driven surrogate forecast model with a training-free variational assimilation module.

The session keynote, given by Takemasa Miyoshi (RIKEN—Japan), connected real time DA to synergistic DA and AI development and the continuing convergence of large-scale computing, rapid-update assimilation and learning systems. Nina Raoult (ECMWF—UK) widened the scope to a data-driven Earth-system model spanning ocean, sea ice, land, waves and hydrology, including questions of coupling strategy and physical consistency across components. Mikko Partio (Finnish Meteorological Institute—Finland) addressed operational usability, showing how frozen core weather models can be extended to additional variables through separate decoders rather than full retraining, with wind-gust prediction as a test case. Claire Merker (MeteoSwiss—Switzerland) presented a regional Alpine forecasting workflow based on high-resolution datasets, real-time inference and time interpolation between forecast steps. Interpretability and physics diagnostics were explicitly discussed by George Craig and Matthias Beylich (both from Ludwig-Maximilians-Universität München—Germany). Probabilistic extremes and regional downscaling appeared in the talks by Thomas Rieutord (Météo-France) and Athul Rasheeda Sathesh (Karlsruhe Institute of Technology—Germany) while the path toward reliable forecasts of extreme event likelihoods through diffusion models was discussed by Peter Manshausen (NVIDIA—USA). TA4 concluded with a recorded open discussion.

**Limitations, opportunities, and challenges.** End-to-end data driven workflows are increasingly considered for operational systems, but questions remain regarding purely data driven approaches versus physics-based progress. Across the session talks, a consistent message was that end-to-end systems are advancing rapidly, but operational credibility depends on more than headline forecast skill. This session presentations explicitly engaged with those questions through



interpretability and “physics of AI” discussions, suggesting recognition that operational trust requires more than forecast skill, e.g. physical consistency, robust uncertainty, and diagnosability. Current challenges include cycling and observation network heterogeneity in observation centric systems, stability under extremes, and meaningful probabilistic verification for generative methods.

A central thread of this session discussions explored the extent to which better reanalyses can be produced by observation-only AI/ML systems by avoiding the systematic errors introduced by physics-based models. Panellists agreed that this overlooks key issues. One is inverse problems and unobserved variables: radiance-to-state inversion is non-trivial, many model variables (and some parts of the Earth system) are not directly observed and physical models still play a central role in inferring full states. Fundamentally, reconstructed Earth system states from reanalyses need to be physically consistent to be useful in Climate Science e.g., for the estimation and attribution of climate trends. Physical consistency is, within approximations, directly enforced in the physics-based models used in reanalysis, but it is only expected to arise as an emergent property in fully observation driven estimation frameworks. Panellists also discussed the “variables question”: operational users require a wide set of products, while some end-users care about a narrow subset (e.g., high-impact variables). The discussion suggested ML may enable direct prediction of user-specific products, potentially bypassing some diagnostic post-processing steps, but that operational systems still provide superior breadth and reliability.

On evaluation of predictive skill, panellists noted that observation-centric systems may show uneven skill across variables (e.g., better surface skill than upper-air fields), and suggested this could reflect both observing-system density and “data-limited” regimes (e.g., fewer independent samples for slowly varying large-scale fields). This prompted discussion of observing-system design: observation-centric ML models may enable new sensitivity analyses that reveal where network gaps limit performance, motivating potentially new design requirements.

Finally, panellists explored why cycling AI/ML models inside DA systems can be difficult due to the need for reliable Jacobians, coherent multivariate correlation structures for ensemble DA, and the role of the model as a strong constraint that propagates both information and covariance through cycling—functions that observation-only systems must reproduce in some form.

**Future directions.** The presentations and group discussion highlighted four priority areas of future development:

1. Clarify “what replaces what”: distinguish observation-to-observation forecasting from full state estimation and articulate how inverse problems and unobserved variables are handled in fully observation-based estimation and prediction.
2. Observation-centric benchmark design: build benchmarks that explicitly test dependence on observing system density, variable observability, and lead time.
3. Observing system co-design: use sensitivity and synthetic-data experiments to assess how observation networks should evolve for end-to-end approaches.
4. DA-compatibility of ML models: prioritise architectures and training methods that support stable cycling, coherent multivariate structure, and robust uncertainty propagation.

**TA5: high-performance and new computing technologies for ML applications in ESOP.** This session was chaired by Nina Raoult (ECMWF—UK) and Diego Jatobá Dos Santos (ESA—Italy). The themes discussed in

this session were related to scaling and profiling ML emulators and next-generation computing paradigms.

**Current ML applications.** Bastian Koller’s (High Performance Computing Center [HLRS] —Germany) opened the session by setting the HPC-AI context, describing how vendor collaboration has reduced peak power consumption from four megawatts to two and how waste heat reuse will be incorporated into next-generation infrastructure, alongside centres such as Large Unified Modern Infrastructure (LUMI) that already operate on entirely green energy. Practical performance engineering for ML forecast models was a strong focus, with talks on scalable GPU-accelerated training of graph neural networks for high-resolution ESOP simulations (Oscar Leung, Hong Kong university and Ye Ha Kim, UCL—UK), improving the performance and scalability of Anemoi (Jan Polster, ECMWF—UK), and optimising Anemoi for higher-resolution data and next-generation systems (Cathal O’Brien, ECMWF—UK). Profiling ML weather forecasting models in Anemoi (Marieke Plessle, ECMWF—UK) demonstrated that existing tools such as EC-PEAK, originally designed for the IFS physical model, could be adapted to ML workloads with modest effort, with the main shift being greater emphasis on GPU utilization, data loading, and memory usage. The Anemoi team reported increasing integration of Triton kernels to better exploit hardware capabilities, and noted the growing influence of hardware-software co-design, exemplified by NVIDIA’s Grace Hopper architecture and the inherent suitability of the attention mechanism for GPU parallelisation. Interest in more disruptive paradigms connected to Quantum Machine Learning appeared in the talks by Federica Torrisi (INGV—Italy), where quantum circuits were embedded within neural network architectures for satellite image classification, with models first simulated in Qiskit before deployment on quantum hardware; and Eleonora Amato (INGV—Italy), where quantum layers were shown to be useful to improve the expressivity of AI/ML models used in super-resolution tasks for geophysical applications.

**Limitations, opportunities, and challenges.** Energy efficiency and resource utilization emerged as central tensions: a system running at 50% utilization was characterised as an “ecological catastrophe”, since idle hardware still draws significant power, yet maximising utilization is constrained by legacy infrastructure and capital availability. The model serving design pattern—separating model logic from the broader workflow to avoid reserving GPUs unnecessarily—is straightforward in cloud environments but remains difficult in traditional HPC setups built around batch schedulers such as Slurm, prompting a case-by-case “concierge service” approach, e.g. at HLRS Stuttgart. CPU/GPU memory offloading was identified as promising for reducing VRAM pressure in large-scale graph neural network training, but creates practical conflicts with high-resolution weather models that require all available CPU memory for data loader processes; offloading dormant node features can consume hundreds of gigabytes of CPU RAM, and balancing VRAM savings against CPU memory availability remains an active area of research. The rapid pace of hardware and software evolution—GPU drivers, PyTorch releases, and profiling tools changing on 6-month to 1-year cycles—requires constant adaptation, though panellists noted that profiling methodology for ML workloads is broadly similar to that for physical models and that even one person working as a side project can make meaningful progress. Quantum machine learning faces its own constraints, including near-zero kelvin cooling requirements, noise management, and prohibitive cost—IBM’s pay-as-you-go access currently charges approximately \$1000 for 10 minutes of quantum compute

time—motivating simulation-first development workflows. The broader discussion highlighted concerns about system fragmentation and the operational burden of supporting increasingly diverse hardware and software stacks.

**Future directions.** The panel identified AI, HPC, and quantum computing as complementary tools in a shared toolbox rather than competitors, at least for the present, with quantum computing expected to follow a trajectory analogous to GPUs a decade ago—beginning as an accelerator to which specific kernels are dispatched before eventually maturing into systems capable of running full workloads. Quantum machine learning was seen as particularly promising for feature extraction that may be intractable classically, with the distinction drawn between quantum computing as a general paradigm and quantum AI/ML as its application at the intersection of ML and quantum computation. Over a 10–15 year horizon, one technology may come to dominate, but panellists anticipated a shift away from general-purpose supercomputers toward domain-specific systems tailored to communities such as Earth observation or engineering. The need for robust offloading and memory management strategies, further integration of physical laws within ML models, and maintaining open access to large-scale computing resources were all identified as priorities, with the overarching responsibility framed as providing these converging technologies in the best possible way to address real-world problems in Earth system science.

## Conclusion

The 5th ML4ESOP workshop edition reflected a community rapidly expanding the scope and ambition of ML in ESOP, from Digital Twin components and EO applications to hybrid operational systems and end to end, observation centric modelling, under tight coupling to a shifting compute landscape. In this development path, practitioners must be increasingly mindful of the operational, societal, and sustainability constraints associated with scaling ML-driven and hybrid applications towards realistic applications.

TA group discussions converged on a pragmatic message: near-term progress will likely be driven by hybrid and reliability-aware approaches, sustained by shared benchmarks and data assets, modernised toolchains, and explicit attention to user trust and operational governance. Longer term, participants anticipated greater co-design among observing systems, ML architectures, DA methodologies, and heterogeneous computing platforms, with an increasing need to quantify potential trade-offs among accuracy, reliability, interpretability, and energy use. This will clearly require an even greater level of collaboration among DA, Observations and ML practitioners, which is specifically what this series of Workshops is designed to foster. In the longer term, AI is expected to become an integral layer of Earth system observation and prediction, not as a replacement for physical modelling and data assimilation, but as a means to make them more adaptive, efficient and information-rich. By combining physical knowledge, uncertainty-aware data assimilation and learning from heterogeneous observations, AI can support more reliable prediction systems, better exploitation of emerging data sources, and more actionable services for weather, climate and environmental risk.

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## References

1. Ebel, P., Schneider, R. & Bonavita, M. 2024 ESA-ECMWF workshop report: current status, progress and opportunities in machine learning for Earth system observation and prediction. *npj Clim. Atmos. Sci.* **7**, 241 (2024).
2. Bonavita, M. et al. 2022 ECMWF-ESA workshop report: current status, progress and opportunities in machine learning for Earth System observation and prediction. *npj Clim. Atmos. Sci.* **6**, 87 (2023).
3. Schneider, R., Bonavita, M. & Geer, A. ESA-ECMWF Report on recent progress and research directions in machine learning for Earth System observation and prediction. *npj Clim. Atmos. Sci.* **5**, 51 (2022).
4. Bonavita, M. et al. Machine learning for Earth system observation and prediction. *Bull. Amer. Meteor. Soc.* **102**, E710–E716 (2021).

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## Author contributions

M.B. and R.S. defined the questions that structured the basis of the manuscript. M.B. and R.S. took the lead in writing the manuscript. All authors contributed to writing their own TA sections and providing critical feedback to the overall manuscript.

## Competing interests

The authors declare no competing interests.

## Additional information

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